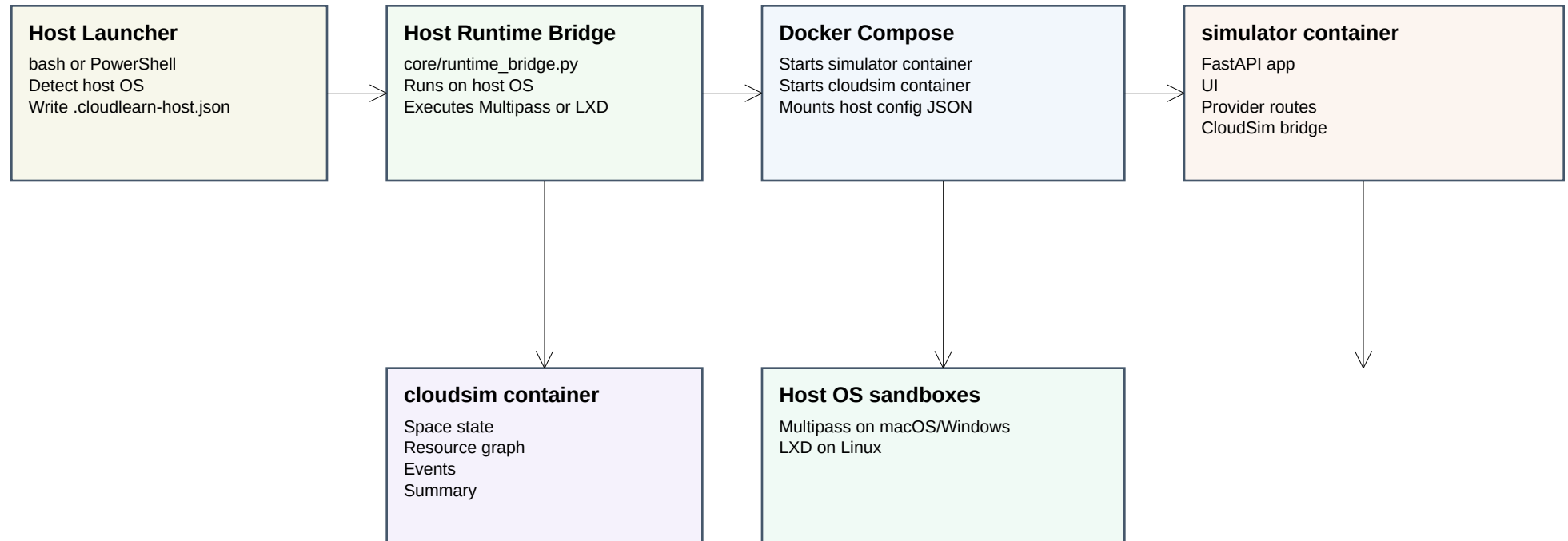


CloudLearn Service Spawn Diagram

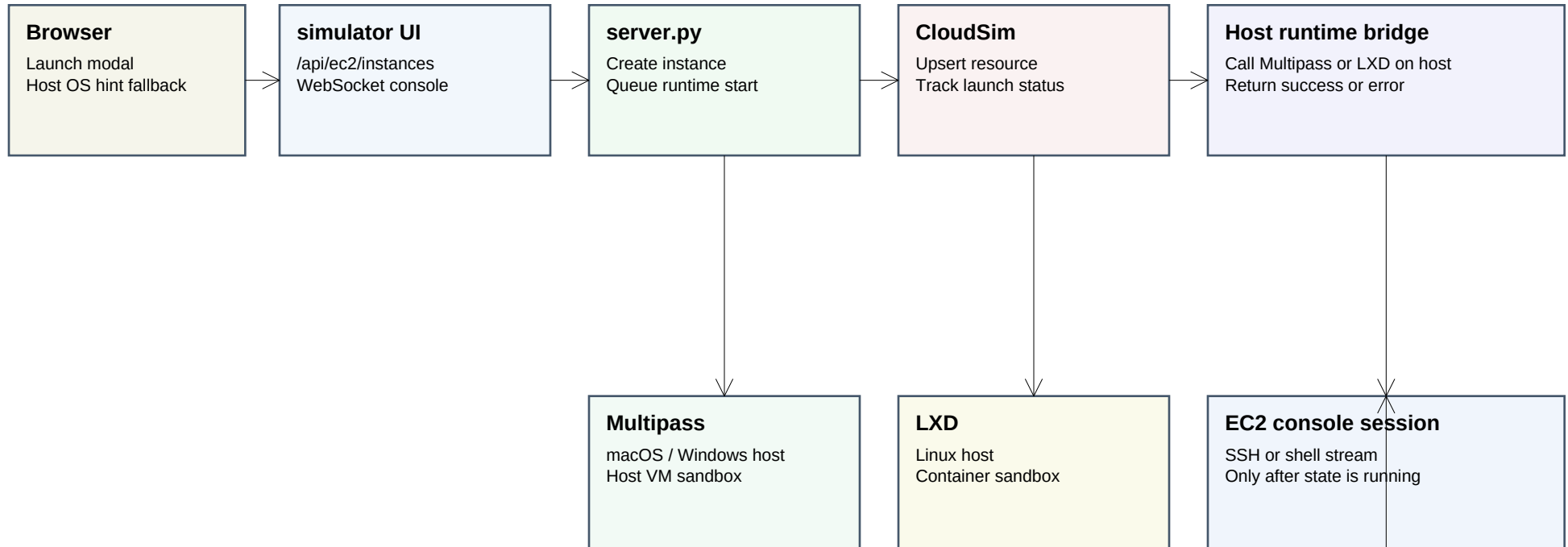
How the simulator starts, where host state lives, and how EC2 sandboxes are launched



Key rule: the simulator container never launches Multipass or LXD directly. It asks the host bridge to do it on the host OS.

EC2 Launch Sequence

The instance is created in the simulator first, then the host bridge attempts to start the real sandbox



Failure mode: if the host bridge is unreachable, the instance stays pending and launch_status becomes error. The UI shows launch failed.

Responsibilities by Layer

This is the clean split that keeps the architecture understandable



Launch states

- pending: the instance record exists but the runtime is not yet confirmed.
- running: the host runtime started successfully and the console can connect. Multipass is used on macOS and Windows.
- launch_status=error: the host bridge could not start the sandbox.
- terminated: the sandbox was removed and the instance is no longer active.

Host bridge checks

- Bridge health must be reachable from the container.
- LXD is used on Linux.
- The browser OS hint is only a fallback; host config JSON is preferred.